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Data and Knowledge-Driven Bridge Digital Twin Modeling for Smart Operation and Maintenance

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Abstract: The rapid expansion of civil infrastructure in China underscores the critical need for advanced solutions to ensure the structural health of aging bridges. This study introduces a novel data and knowledge-driven digital twin modeling (DK-DTM) framework designed to enhance the safe and efficient operation and maintenance (O&M) of bridges. Such a system should be capable of (1) monitoring structural dynamics in real time, (2) capturing spatiotemporal details and changes (e.g., defects and deformations), (3) analyzing structure deterioration patterns, (4) predicting structure failure risks, and (5) generating optimal maintenance and repair actions for ensuring structural safety. Previous studies have developed advanced sensing techniques and robust artificial intelligence algorithms for capturing and analyzing bridge health conditions. However, most existing techniques and algorithms heavily rely on high-quality data, which are difficult to obtain during bridge O&M. This raises the critical question of how to incorporate expert knowledge together with data-driven tools to establish a trustworthy DT for bridge O&M. This study presents the DK-DTM framework, which uniquely integrates multi-source data collection, spatiotemporal modeling, and expert knowledge reasoning. By combining these components, the framework supports smart structural health assessments of bridges, enabling comprehensive monitoring, prediction, and decision-making for efficient maintenance. The spatial and temporal models provide real-time data, while the expert knowledge model functions as an automated evaluation tool for structural health assessment. The results demonstrate that the proposed DK-DTM framework significantly enhances the accuracy and efficiency of O&M processes for aging bridges, addressing key gaps in existing digital twin methodologies.

Keywords: digital twin model; structural health assessment; expert knowledge

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1. Introduction

Since the 1970s, the number of bridges in China has expanded rapidly. According to the 2023 Statistical Bulletin of Transportation Industry Development, there were 1,079,300 highway bridges nationwide by the end of 2023. The rapid increase in the number of bridges, combined with higher vehicle density and load demands, has created significant challenges in ensuring their safe and efficient operation and maintenance (O&M) [1]. Aging bridges face structural issues such as cracking, water seepage, and damage to expansion joints, requiring advanced solutions for long-term monitoring and maintenance. In the United States alone, approximately USD 20.5 billion is needed annually to address the

escalating deterioration of bridges [2]. Therefore, a smart bridge management system is necessary for ensuring safe and efficient O&M of aging bridges.

Traditional visual inspections of bridges are labor-intensive and rely heavily on subjective judgments, often leading to inconsistent and error-prone results. This highlights the need for automated solutions that can improve both the accuracy and the efficiency of bridge health monitoring. Recent developments in computer vision techniques and image processing algorithms show potential in bridge inspection and structural health monitoring (SHM) of bridges [3]. Despite their widespread applications, computer vision techniques and image processing algorithms still face challenges, including noise interference, limited sensitivity, the complexity of setting appropriate alarm thresholds, and sensor reliability. Especially when dealing with complex bridge structures, these factors may result in missed detections, false positives, or system malfunctions, thereby affecting the overall effectiveness and reliability of the bridge health monitoring system [4]. It is therefore necessary to establish a smart O&M system that can monitor structural dynamics and capture spatiotemporal details for analyzing structure deterioration patterns and assessing structure failure risks.

A trustworthy BDT enables seamless integration between physical and virtual environments by providing an accurate digital representation of physical structures, utilizing real-time data, and facilitating closed-loop interactions between the virtual and physical bridges [5]. The integration of virtual and physical systems overcomes limitations in time, space, cost, and safety, enabling comprehensive real-time assessments and predictive maintenance for bridge structures. Therefore, DT technology can be applied in the field of bridge SHM, significantly improving its operational efficiency. In fact, DT technology has been extensively applied in various fields, including manufacturing, aerospace, transportation, healthcare, and energy. DT technology has gained substantial attention and recognition from academic institutions, government agencies, and industry experts both domestically and internationally [6].

The current digital twin modeling (DTM) is typically constructed using collected data to construct simulation models. However, relying solely on data-driven DT models makes it difficult to comprehensively monitor and provide early warnings for the health status of complex bridge structures. This study proposed a DK-DTM framework for supporting smart structural health assessments of bridges. The proposed framework contains (1) data collection and processing of bridge health condition data from multiple sources and (2) spatiotemporal modeling of BDTs. The proposed DK-DTM framework uniquely integrates spatiotemporal modeling with expert knowledge reasoning to provide dynamic early warnings for potential structural issues. This comprehensive approach ensures accurate monitoring, predictive analysis, and enhanced decision-making for bridge O&M. As a result, the proposed framework significantly enhances the efficiency and safety of bridge O&M.

The organization of the paper is as follows. Section 2 provides a systematic review for synthesizing previous studies on SHM and DT and identifying knowledge gaps. Section 3 illustrates a DT framework and modeling process for bridge O&M. Section 4 discusses the data collection and processing methods essential for building the DTM, including 2D image data, 3D point cloud data, sensor data, and expert knowledge. Section 5 presents the multi-source heterogeneous data fusion model, which integrates spatiotemporal and expert knowledge models to enable a comprehensive assessment of structural health. Section 6 provides a discussion of the proposed model's strengths and limitations compared with traditional approaches. Finally, Section 7 concludes the paper by summarizing the main contributions and proposing directions for future research.

2. Literature Review

2.1. Structural Health Monitoring

Over the past few decades, driven by advancements in sensing technology, data processing, and artificial intelligence (AI), bridge structural health monitoring (SHM) technology has experienced significant advancements. In the past, SHM mainly relied on manual inspections and non-destructive testing (NDT) techniques, such as acoustic emission, X-ray diffraction, and interferometry [7]. However, these methods have certain limitations, particularly in addressing complex bridge structures, as well as in real-time monitoring and accuracy [8].

Recent developments have underscored the potential of big data and AI for transforming SHM, particularly in terms of interpreting vast amounts of data to facilitate effective condition assessment [9]. The advent of digital technologies has facilitated a transition to more sophisticated, automated SHM systems. Long-span bridges worldwide, such as the Tsing Ma Bridge in Hong Kong and the Akashi Kaikyo Bridge in Japan, have been equipped with long-term monitoring systems, enabling continuous data collection and damage detection [10]. These systems utilize networks of sensors, including strain gauges, fiber optics, and wireless sensors, to monitor key parameters such as vibration, strain, and temperature [11]. The integration of robotics, including ground, wall-climbing, and flying robots, has further enhanced these monitoring capabilities by enabling access to hard-to-reach areas and increasing the efficiency of data collection [12]. The Internet of Things (IoT) has also contributed to the transition from traditional wire-based systems to wireless, real-time SHM, enhancing data accessibility and system efficiency [13]. For instance, Serlenga et al. (2021) demonstrated the importance of integrating diverse sensing techniques, such as seismic and electromagnetic methods, to monitor the Gravina Bridge in Matera, Italy [14]. Their study showcased how multi-source data fusion enables a holistic analysis of structural behavior, improving the reliability of SHM systems while addressing challenges associated with merging heterogeneous data types. While significant progress has been made, several challenges persist in SHM, particularly regarding data interpretation, model accuracy, and system reliability. Traditional data-driven methods, although widely used, often struggle to accurately capture the complexity of bridge behavior under varying environmental and operational conditions [15]. Heterogeneous data from diverse sensor types further complicate SHM processes, requiring advanced data fusion techniques to improve model accuracy and system reliability [9]. In this context, the evaluation of safety formats for non-linear finite element analysis offers valuable insights. For instance, Tur, Viktar V. et al. (2020) critically analyzed widely used methods such as the partial factor method, global resistance methods, and probabilistic approaches, highlighting significant uncertainties and statistically vague formulations in current structural codes [16]. These findings emphasize the need for advanced methodologies to address variability in structural responses and improve the reliability of safety evaluations, particularly for aging bridge infrastructure. Numerical modeling approaches, including non-linear analysis methods, further contribute to the field by addressing uncertainties in structural performance assessment. As noted by De Matteis et al. (2023), non-linear analysis frameworks enable accurate simulations of bridge behavior, providing robust methodologies for assessing safety under complex conditions [17].

The integration of big data and AI technologies into SHM offers significant potential to address these gaps. Machine learning algorithms, particularly those used for vibration- and image-based SHM, have exhibited considerable potential in damage detection, pattern recognition, and predictive analysis [18]. However, many of these systems are still in the developmental phase, and challenges remain, such as managing heterogeneous data from different sensor types and improving the reliability of real-time predictions [19].

In light of these challenges, the DK-DTM presents a viable solution. By incorporating expert knowledge into the rule engines and data interaction mechanisms, this model improves both the accuracy and predictive capabilities of SHM systems. The integration of real-time data with expert knowledge in DT technology enables a more holistic monitoring framework, significantly improving the efficiency and safety of bridge operation and maintenance.

2.2. Digital Twin Technique

The concept of the DT, first proposed by Michael Grieves, was initially applied in product lifecycle management (PLM) [20]. NASA pioneered the application of DT technology in the aerospace field in 2010. As technologies such as the Internet of Things (IoT), artificial intelligence (AI), and big data advance, DT technology has gradually expanded into more industries [21].

DT technology has become widespread across sectors such as manufacturing, healthcare, aerospace, and construction infrastructure [22]. In manufacturing, the adoption of DTs has been key in facilitating cyber-physical integration, optimizing production efficiency, and enabling predictive maintenance through real-time data analysis [23]. Leveraging data-driven approaches, DTs have transformed smart manufacturing, allowing for enhanced decision-making and streamlined operations across the product lifecycle [24]. In healthcare, it supports personalized treatment and public safety management [20]. In construction and infrastructure, especially in bridge SHM, DT technology facilitates real-time bridge monitoring and predictive maintenance [25]. Digital twins rely on technologies such as IoT, big data analytics, cloud computing, and AI [26]. Modern DTM not only encompasses physical and virtual entities but also integrates data, services, and networks, enhancing interaction between the physical and virtual [27]. Furthermore, machine learning algorithms and big data analytics optimize system performance and predict future trends and failures [28]. DTM is used throughout the infrastructure's lifecycle, offering a dynamic and holistic view of its performance and condition. Studies such as that by Liu et al. (2020) demonstrate the potential of DT-based SHM systems for large infrastructure, highlighting the importance of real-time data integration in bridge management [29]. In bridge monitoring, technologies like machine vision, edge computing, and big data fusion enhance real-time monitoring and data processing efficiency for bridges. Machine vision and multi-source data fusion technologies enable effective monitoring of bridge traffic loads and structural health, facilitating timely detection of potential issues and preventive maintenance [30]. It enables multi-dimensional monitoring of bridge structures, allowing timely, data-driven maintenance decisions [31].

Despite significant advances in many fields, DT technology still faces numerous challenges in practical applications. Constructing and maintaining DTM requires extensive data support, with high costs for data collection, processing, and storage, particularly in the monitoring of large infrastructure such as bridges [28]. Moreover, as scenarios grow more complex, creating a multi-dimensional, multi-scenario DTM presents substantial technical and resource challenges [27]. For infrastructure such as bridges, achieving multi-source data fusion and processing, along with building a high-precision, automated DTM, remains a central focus of current research.

3. A Data and Knowledge-Driven Digital Twin Framework for Smart Bridge O&M

This study proposed a BDT model driven by both data and expert knowledge to address the limitations of traditional models concerning accuracy and intelligence. The model comprises three main components: the spatial model, temporal model, and expert

knowledge model. These models work in unison to facilitate real-time health monitoring and generate early warnings for bridges, such as alerts for structural damage, deformation, or other critical anomalies.

The spatial and temporal models utilize high-resolution cameras, light detection and ranging (LiDAR), and various sensors to collect multi-source data from the bridge structure, including 2D image data, 3D point cloud data, and sensor time series data. The spatial model primarily describes the bridge's spatial characteristics through the modeling of 2D image and 3D point cloud data, providing an accurate spatial representation of the bridge structure. The temporal model focuses on the bridge's physical characteristics, capturing changes in internal physical states and long-term developmental trends under load.

The expert knowledge model consolidates insights from expert interviews, workshops, industry standards, and historical case analyses. This aggregated knowledge is then standardized and structured into an expert knowledge base, which integrates extensive expertise in bridge maintenance and inspection to offer theoretical support and practical guidance for evaluating and assessing bridge conditions. The spatial and temporal models together form the data-driven module, which is responsible for collecting and processing real-time information and developmental trends of the bridge structure. These data are then transmitted to the expert knowledge-driven evaluation module. The evaluation module automatically analyzes and assesses the data by incorporating knowledge from the expert knowledge base, providing a comprehensive assessment of the bridge's condition and predicting its health status. Thus, the DTM, combining data-driven and knowledge-driven approaches, forms a powerful bridge health monitoring and early warning system. The DK-DTM framework shown in Figure 1.

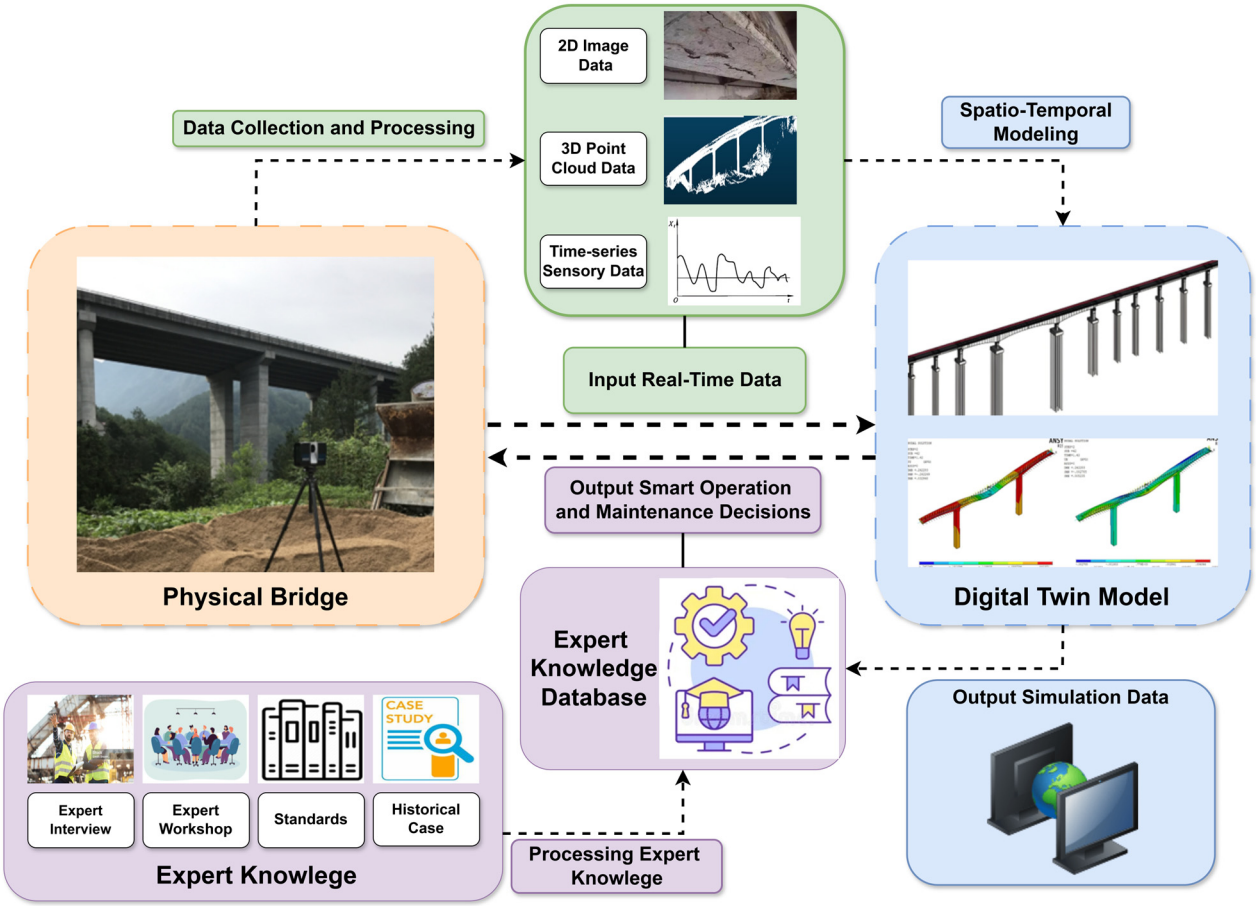


Figure 1. The data and knowledge-driven digital twin modeling.

4. Spatiotemporal Data Collection and Processing for Bridge Digital Twin

This chapter systematically presents the key methods for data and expert knowledge collection and processing that are critical to the development of DTM of bridges. Accurate data collection and efficient processing are paramount for real-time monitoring and the early detection of structural deterioration. First, the chapter introduces techniques for 2D image data collection and the application of the YOLOv8 model to automatically detect surface defects, such as cracks and spalling. Next, the chapter discusses the application of LiDAR and related technologies to acquire 3D point cloud data, with the application of PointNet algorithms to reconstruct the three-dimensional geometric model of the bridge, which is fundamental for spatial analysis. Then, the chapter presents the collection and processing of sensor time series data, emphasizing the monitoring of key physical parameters, including stress, strain, and environmental conditions, to enable real-time structural assessment. Finally, this chapter presents the process of collecting and processing expert knowledge, establishing an expert knowledge base through fuzzy logic. The data processing procedure of the DTM is shown in Figure 2.

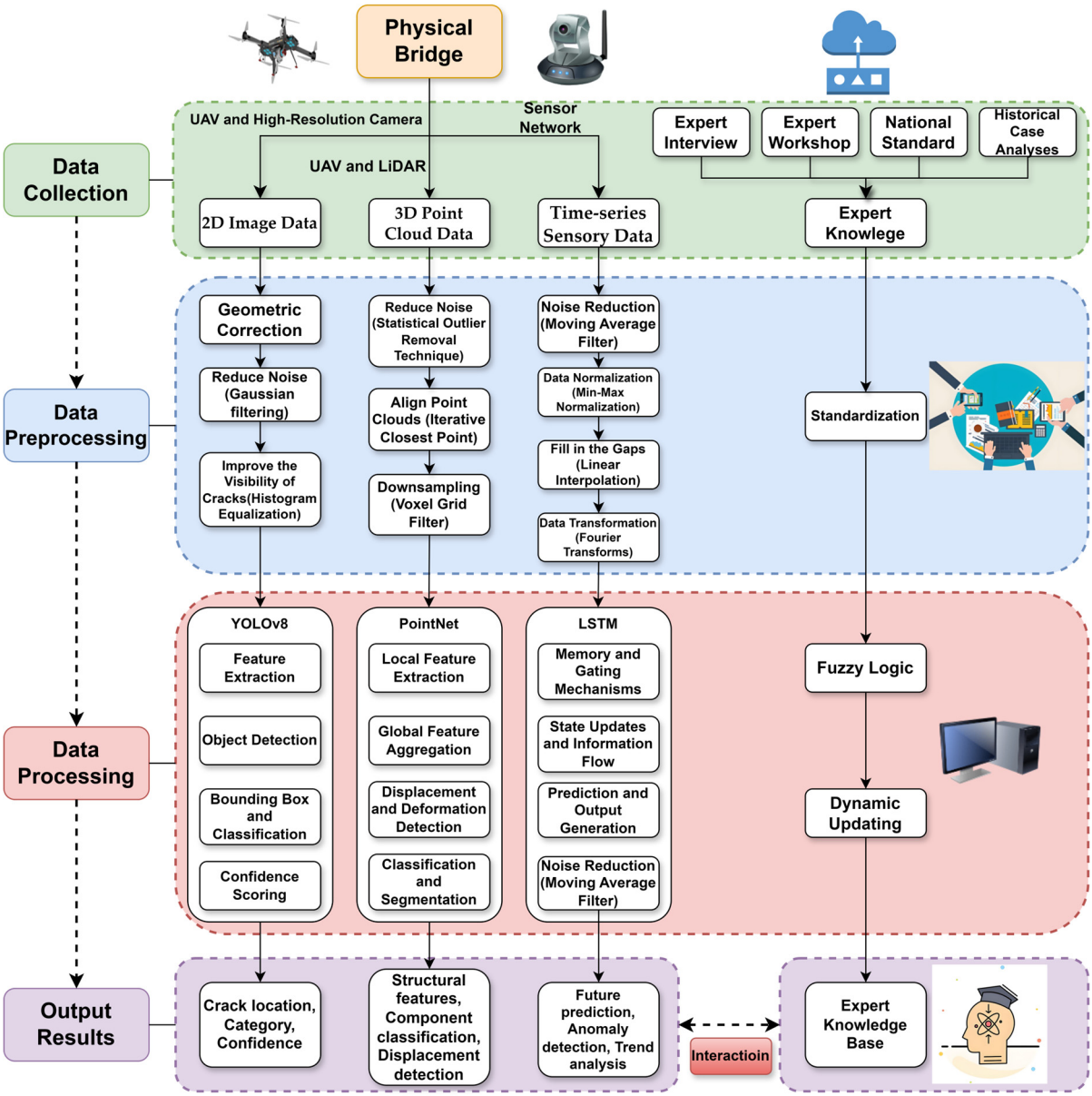


Figure 2. The data processing procedure of the digital twin model.

4.1. The 2D Image Data Collection and Processing

4.1.1. The 2D Image Data Collection

The accurate collection of 2D image data is essential for detecting surface deterioration in bridge structures, including cracks and spalling. In this study, unmanned aerial vehicles (UAVs), equipped with high-resolution cameras, were used to capture detailed images of the bridge surfaces. UAVs can ensure comprehensive coverage of all areas of the bridge, particularly offering considerable advantages in accessing regions that are difficult to reach by other means, such as the undersides of bridges and elevated sections. This capability is especially valuable for large-scale bridges, where traditional methods struggle to achieve comprehensive coverage of all areas. Moreover, this method ensures that the collected images are of high enough resolution to detect even fine surface cracks. The use of UAVs substantially reduces the time and labor typically required for manual inspection while improving the overall precision and efficiency of the data collection process. The 2D image data collection is shown in Figure 3.

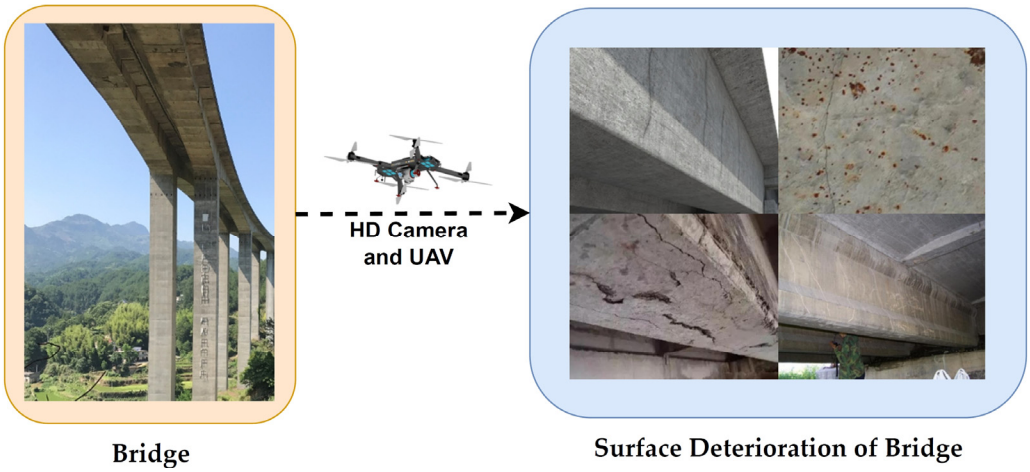


Figure 3. Image data collection of bridge surface defects.

4.1.2. The 2D Image Data Preprocessing

After image collection, preprocessing was necessary to ensure the data were suitable for accurate detection. Given the variability in UAV flight angles, geometric distortion was corrected using a perspective projection model [32]. This correction ensured that the spatial relationships in the images precisely represented the true geometry of the bridge. The corrected image coordinates (x', y') are given by Formula (1):

$$x' = \frac{f \cdot x}{z} \quad y' = \frac{f \cdot y}{z} \tag{1}$$

where x', y' are the pixel coordinates in the image; x, y represent the real-world coordinates on the bridge surface; f is the focal length of the camera; and z is the distance from the camera to the bridge surface.

Both optical and infrared images underwent alignment to ensure consistency across datasets, enabling the joint analysis of surface and subsurface conditions. After geometric correction, Gaussian filtering was applied to reduce noise [33]. Gaussian filtering enhanced image smoothness while retaining essential details such as crack edges and is defined as in Formula (2):

$$I'(i,j) = \sum_{m=-k}^{k} \sum_{n=-k}^{n} I(i+m, j+n) \cdot G(m,n) \tag{2}$$

where $I(i,j)$ is the intensity of the pixel at coordinates (i,j) in the original image, $I'(i,j)$ is the intensity of the same pixel after applying the filter, $G(m,n)$ represents the Gaussian kernel, and k defines the size of the filter window.

To further improve the visibility of cracks in the optical images, histogram equalization was applied [34]. This method adjusted the intensity distribution of the pixels, improving the contrast and making fine cracks more easily detectable. In the infrared images, thermal contrast enhancement was used to emphasize temperature variations that may indicate structural weaknesses.

4.1.3. The 2D Image Data Processing Using YOLOv8

After preprocessing, the 2D image data were analyzed using the You Only Look Once, Version 8 (YOLOv8), model to automatically detect cracks [35]. YOLOv8 is a highly efficient and rapid algorithm that performs both object classification and localization in a single step. It facilitates the identification of cracks on the surface of the bridge by predicting their location and size, as shown in Figure 4.

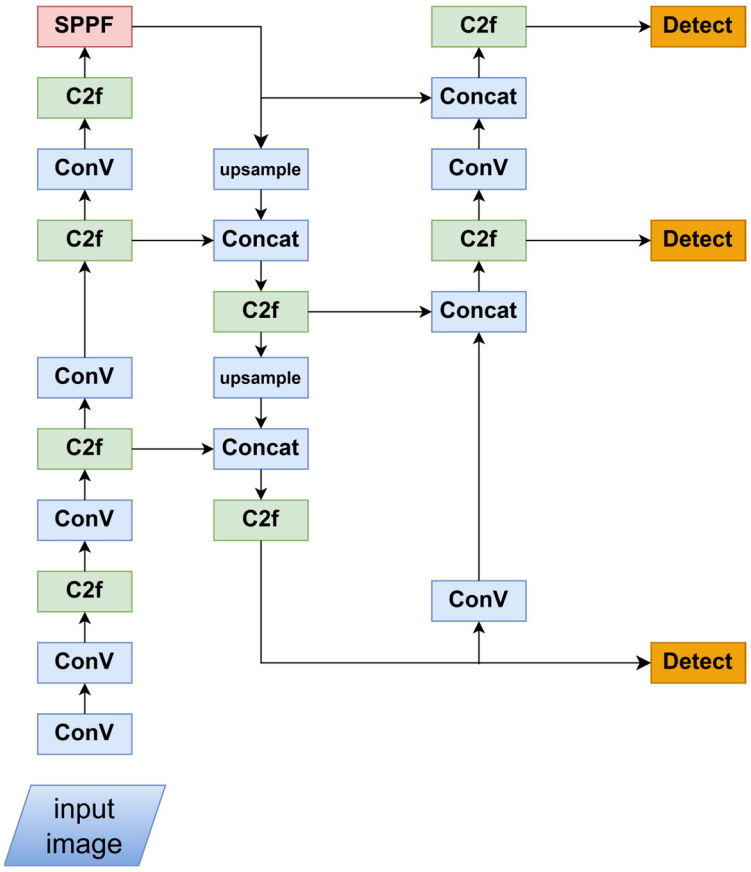


Figure 4. Schematic representation of the YOLOv8 architecture.

For each detected crack, YOLOv8 predicts a bounding box, which includes the center coordinates (b_x, b_y), the width b_w, and the height b_h of the box. These values are calculated based on the model's predictions and the position of the grid cell in the image. The formula for calculating the center of the bounding box is provided in Equation (3):

$$b_x = p_x + c_x\,,\; b_y = p_y + c_y \tag{3}$$

where p_x and p_y are the predicted offsets within the grid cell, and c_x and c_y are the coordinates of the top left corner of the grid cell.

The width and height of the bounding box are also predicted by the model. These values determine the size of the crack and are calculated as follows according to Formula (4):

$$b_w = p_w + e^{t_w}, b_h = p_h + e^{t_h} \quad (4)$$

where p_w and p_h are the predefined anchor box dimensions (representing average crack sizes), and t_w and t_h are the adjustments predicted by the model.

In addition to localizing the cracks, YOLOv8 also classifies them. The model assigns a confidence score to each detected crack, indicating the model's certainty that the detected object is indeed a crack. This allows for more accurate and reliable detection results. The YOLOv8 model's architecture ensures that detection is both rapid and precise, making it suitable for real-time monitoring applications in structural health assessments.

4.2. The 3D Point Cloud Data Collection and Processing

The use of 3D point cloud data is essential for spatial geometric modeling and deformation monitoring of bridge structures. By utilizing LiDAR technology, high-density point cloud data offer a precise and comprehensive representation of the bridge's three-dimensional geometry, enabling the accurate identification of localized structural damage and the assessment of overall deformation. This section outlines the methods for 3D point cloud data collection, data preprocessing, and the application of the PointNet model for processing the data, as shown in Figure 5.

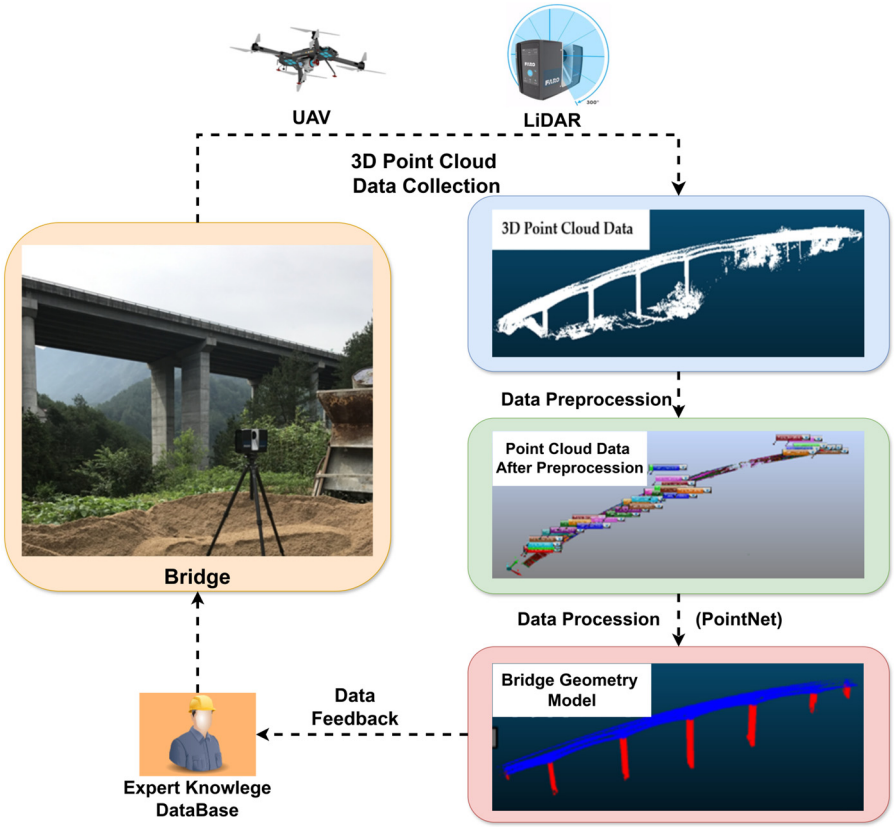


Figure 5. The 3D point cloud data collection and processing [36–38].

4.2.1. The 3D Point Cloud Data Collection

The acquisition of high-precision three-dimensional (3D) geometric data is critical for bridge structural assessments, and LiDAR technology is widely employed for this purpose. LiDAR sensors emit laser pulses and calculate distances to surfaces by measuring the time of pulse returns, producing high-density point cloud data. Depending on the size

and characteristics of the bridge, two primary methods are utilized for data acquisition: UAV-mounted LiDAR systems and stationary rack-mounted laser scanners.

For large-span bridges, such as suspension or cable-stayed bridges, UAV-mounted LiDAR systems are deployed to capture detailed geometric data. These systems enable comprehensive multi-angle scans, particularly for hard-to-access areas, including the underside of decks and tall piers [31]. Flight paths are carefully designed using pre-assigned coordinates to ensure complete coverage of critical structural components, such as piers, decks, and girders. The flight altitude and scan frequency are calibrated to meet the resolution requirements of complex structures. Additionally, UAV-mounted systems are equipped with GPS and inertial measurement units (IMUs), which provide accurate positioning and orientation data to ensure spatial alignment of the collected point cloud.

For smaller or medium-sized bridges, particularly those located in urban environments, stationary 3D laser scanners mounted on racks are typically employed. These scanners are strategically placed around the bridge to capture precise geometric details without requiring UAVs. Rack-mounted systems are particularly advantageous in settings where UAV operation may be constrained by regulatory or environmental factors, such as in densely populated urban areas.

The data collection process generates extensive high-density point cloud datasets that capture key geometric features of the bridge, including cracks, surface irregularities, and localized damage. To produce a unified 3D model, point cloud data from different perspectives are aligned using advanced registration algorithms, ensuring spatial consistency and completeness. This processed point cloud data form the foundation for subsequent applications, including geometric modeling, deformation analysis, and structural health monitoring.

4.2.2. The 3D Point Cloud Data Preprocessing

Following the collection of 3D point cloud data, preprocessing is essential to improve data quality and prepare them for further analysis. The raw data often contain noise and redundant points, which need to be removed using noise removal algorithms (e.g., k-nearest neighbor, KNN) before further processing of the data. In addition, point cloud data obtained from different perspectives need to be aligned using control points, alignment functions or algorithms (e.g., iterative closest point (ICP) algorithms). To ensure accurate geometric modeling and reduce computational complexity, the following preprocessing steps are applied.

First, noise caused by environmental factors, such as lighting conditions or sensor inaccuracies, must be removed. A statistical outlier removal technique is used to address this issue [39]. This method calculates the average distance between each point and its k-nearest neighbors to identify points that deviate significantly from their neighborhood, as shown in Equation (5):

$$\overline{d_i} = \frac{1}{k} \sum_{j=1}^{k} d_{ij} \tag{5}$$

where $\overline{d_i}$ represents the average distance of point k-nearest neighbors, k is the number of nearest neighbors considered for each point, d_{ij} is the Euclidean distance between point i and its neighboring point j.

Points with average distances $\overline{d_i}$ exceeding a predefined threshold are classified as noise and subsequently removed from the dataset. This step enhances the clarity and quality of the data.

Second, when multiple scans of the same structure are taken from different viewpoints, the individual point clouds must be aligned. The iterative closest point (ICP) algorithm is employed to align point clouds by minimizing the Euclidean distance between

corresponding points [40]. The objective function minimized by ICP is expressed as in Formula (6):

$$E(R,T) = \sum_{i=1}^{N} \| p_i^{source} - \left(R \cdot p_i^{target} + T \right) \|^2 \quad (6)$$

where $E(R,T)$ is the objective function representing the total squared distance between corresponding points in the source and target point clouds, N denotes the total number of points in the point clouds, p_i^{source} is the position vector of the point in the i-th point in the source point cloud, p_i^{target} is the position vector of the corresponding i-th point in the target point cloud, R is the rotation matrix aligning the two point clouds, and T is the translation vector shifting the point clouds.

This process optimizes the rotation R and translation T to ensure that the point clouds are spatially consistent, forming a complete 3D representation of the structure.

Finally, to reduce the computational burden while retaining the essential geometric characteristics of the data, a voxel grid filter is applied for downsampling [41]. This method divides the point cloud into a three-dimensional grid of voxels and replaces all points within each voxel with their centroid. The centroid C_v of a voxel V is calculated as in Formula (7):

$$C_v = \frac{1}{|V|} \sum_{i=1}^{|V|} p_i \quad (7)$$

where C_v is the centroid of voxel V, $|V|$ is the number of points contained within voxel V, and p_i represents the position vector of the i-th point in the voxel.

This step significantly reduces the number of points while preserving the overall geometric structure of the bridge, facilitating more efficient processing in the subsequent stages of analysis.

By completing these preprocessing steps—noise removal, point cloud alignment, and downsampling—the 3D point cloud data are optimized for accurate geometric modeling and structural health monitoring.

4.2.3. The 3D Point Cloud Processing Using PointNet

Once the 3D point cloud data have been preprocessed, they are further processed with the PointNet model, a neural network specifically designed for handling unstructured point cloud data [42]. PointNet operates by applying multi-layer perceptrons (MLPs) to each point individually, allowing for the extraction of both local and global features from the data. This unique capability makes PointNet particularly effective in identifying and quantifying geometric deformations in bridge structures, as illustrated in Figure 6.

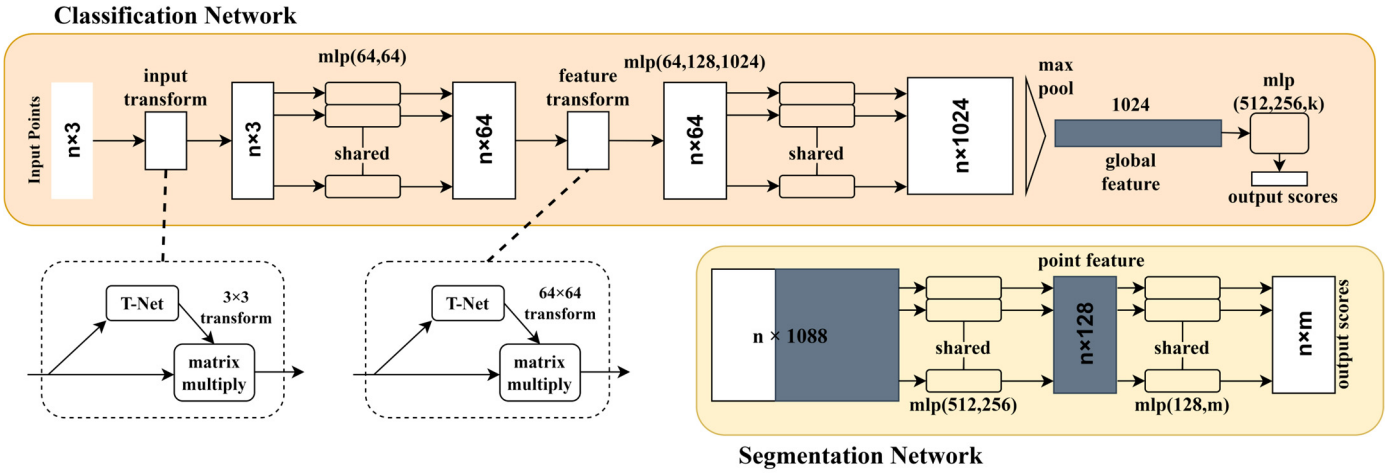


Figure 6. Schematic representation of the PointNet model.

The process begins with local feature extraction for each point $p_i = (x_i, y_i, z_i)$. PointNet applies a sequence of MLP layers, where each layer transforms the point's feature representation. The transformation at the k-th layer is governed by Formula (8):

$$h_i^{(k)} = \sigma\left(W_k h_i^{(k-1)}, +b_k\right) \tag{8}$$

where $h_i^{(k)}$ represents the feature vector after the k-th layer, W_k is the weight matrix of the k-th layer, b_k is the bias term at the same layer, and σ is the non-linear activation function applied to the output.

This process generates a high-dimensional feature vector for each point, effectively capturing local geometric characteristics that are critical for monitoring structural health.

Next, global feature aggregation is performed to summarize the entire structure's geometric shape. This is achieved using a max-pooling operation, which extracts the most prominent features from all the points. The global feature vector g is defined as in Formula (9):

$$g = \max_{i=1}^{N} \{h_i\} \tag{9}$$

where σ is the global feature vector that captures the overall geometry, N is the total number of points in the point cloud, and h_i represents the local feature vector of the i-th point.

This aggregation ensures that the model captures the bridge's overall structural properties, regardless of the number of points, by focusing on the most significant features across the dataset.

For structural deformation monitoring, PointNet enables the calculation of point displacements over time. By comparing the current and reference point clouds, the displacement of each point is calculated as in Formula (10):

$$d_i = \| p_i^{current} - p_i^{reference} \| \tag{10}$$

where d_i is the Euclidean distance representing the displacement of point i, $p_i^{current}$ is the position of point i in the current point cloud, and $p_i^{reference}$ is the position of point i in the reference point cloud.

By analyzing the computed displacements, localized structural deformations, such as cracks or shifts, can be identified, providing valuable insights into the integrity of the bridge.

Finally, component classification and segmentation allow for the differentiation between various structural elements of the bridge, such as piers, decks, and girders. The classification process aims to minimize the cross-entropy loss as in Formula (11):

$$\mathcal{L}_{class} = -\sum_{i=1}^{N} y_i \log \hat{y}_i \tag{11}$$

where $\mathcal{L}_{class}$ is the cross-entropy loss used to evaluate the classification performance, y_i represents the true class label of point i, and $\hat{y}_i$ is the predicted probability of point i belonging to a particular class.

This segmentation enables the system to monitor specific components of the bridge in a more targeted manner, allowing for early detection of issues in distinct areas, such as cracks in the piers or deformation in the girders.

By leveraging PointNet's ability to process unstructured point clouds, this approach supports real-time monitoring of structural deformations. This capability is crucial for providing early warnings in bridge health management systems.

4.3. Time Series Sensory Data Collection and Processing

4.3.1. Sensor Deployment and Data Acquisition

In BDT modeling, an effective sensor network is essential for acquiring high-quality data that accurately reflect the condition of the bridge [43]. Each type of sensor plays a distinct role in capturing specific aspects of the structure's behavior, and their placement must be carefully planned to ensure comprehensive monitoring.

Accelerometers are used to capture the dynamic responses of the bridge, particularly vibrations induced by traffic loads, wind, and environmental forces [44]. These sensors are typically installed on the main girders, deck, and critical load-bearing elements. Their placement is designed to monitor vibrations across various sections of the structure, providing a comprehensive view of the bridge's dynamic behavior. The data collected by accelerometers are essential for identifying modal parameters, such as natural frequencies and mode shapes, which help assess the structural integrity and detect abnormal vibrations that may indicate damage or degradation.

Strain gauges are installed at key structural locations, such as piers, abutments, and support beams, where significant stress is expected. These sensors continuously monitor the strain in the bridge components, allowing engineers to measure how the structure responds to loads over time. The data collected reflect the bridge's deformation under varying loads and provide insights into localized stress concentrations, which are critical for identifying potential fatigue and material failure. Strain data are fundamental for long-term health assessments, helping to detect early signs of structural weakening.

Temperature sensors are distributed across the bridge to monitor environmental and material temperatures, particularly in areas prone to significant thermal expansion and contraction [45]. These sensors are often placed on the deck, beams, and supports where temperature-induced stress may occur. The data collected are essential for adjusting other sensor measurements, such as strain and displacement, as temperature variations directly affect these readings. Accurate temperature data are crucial for distinguishing between temperature-driven structural changes and true material deformations, ensuring the reliability of time series analysis.

Global Navigation Satellite System (GNSS) sensors are placed on the bridge's key components, such as the deck and towers, to measure displacements and monitor large-scale deformations [46]. These sensors are particularly useful for tracking movements due to settling, shifting foundations, or long-term environmental effects. The GNSS data enable the monitoring of slow, cumulative deformations that can impact the stability of the bridge over time. By providing precise measurements of large-scale shifts, GNSS sensors are essential for evaluating overall structural stability.

The data collected from these sensors are time-stamped and synchronized to form comprehensive time series datasets. These datasets undergo further preprocessing to denoise and standardize the data for analysis. Once preprocessed, the data are suitable for advanced time series modeling, such as long short-term memory (LSTM) analysis, which is used to predict structural changes and potential failure points based on historical trends, as shown in Figure 7.

4.3.2. Sensor Data Preprocessing

Preprocessing time series data is an essential step in DTM as it ensures that raw sensor data are preprocessed, reliable, and suitable for advanced analysis. Given that sensor outputs often contain noise, missing records, and inconsistencies, it is necessary to address these issues to enhance the data's usability for predictive models, such as LSTM networks. Therefore, the key preprocessing tasks include noise reduction, data normalization, imputation of missing data, and transforming the data into a format appropriate for model

input. These steps collectively ensure data integrity and accuracy, allowing for meaningful insights into the structure's behavior over time.

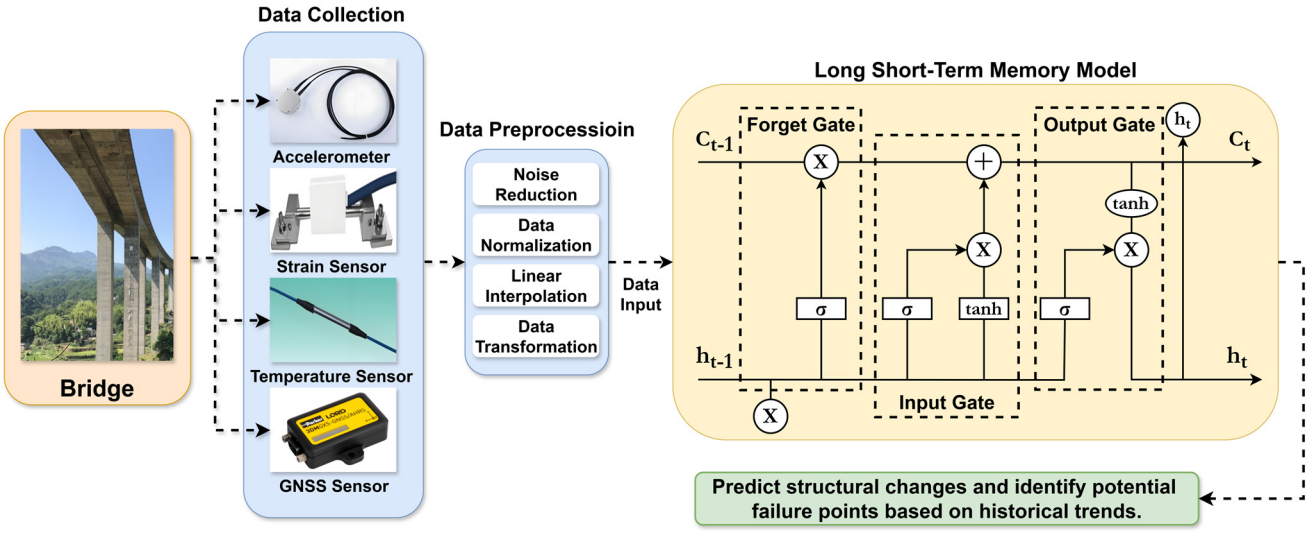


Figure 7. Time series sensory data collection and processing.

The first step in preprocessing is noise reduction as raw data often contain unwanted variations that can obscure the true structural signals. Noise typically arises from environmental factors, equipment calibration errors, or mechanical disturbances. To mitigate these, one of the simplest techniques is a moving average filter [47]. For instance, the smoothed value $\hat{x}_t$ at time t can be computed over a window w as in Formula (12):

$$\hat{x}_t = \frac{1}{w} \sum_{i=t-w+1}^{t} x_i \tag{12}$$

where $\hat{x}_t$ represents the smoothed value at time t, x_i represents the observed value at time i, and w is the size of the smoothing window. This step smooths out short-term fluctuations.

Once noise has been reduced, the next step is data normalization. Since various sensors measure different physical quantities on different scales, normalization aligns all sensor outputs to a common range. This is crucial, especially when feeding the data into machine learning algorithms, which expect features to be on comparable scales. A commonly used normalization method is min-max normalization, which transforms the data into a specified range, typically [0, 1], as in Formula (13):

$$x_t{}' = \frac{x_t - min(x)}{max(x) - min(x)} \tag{13}$$

where $x_t{}'$ is the normalized value at time t, and $min(x)$ and $max(x)$ are the minimum and maximum values in the dataset, respectively.

Following normalization, it is important to handle missing data, which often occur due to sensor malfunctions, transmission errors, or environmental factors. Ignoring missing values could distort the overall analysis; hence, imputation techniques are employed to fill in the gaps. A basic approach is linear interpolation, which estimates missing values based on neighboring observations. If data are missing at timestamp t_m, the imputed value x_{t_m} is computed as in Formula (14):

$$x_{t_m} = x_{t_{m-1}} + \frac{t_m - t_{m-1}}{t_{m+1} - t_{m-1}} \cdot \left(x_{t_{m+1}} - x_{t_{m-1}}\right) \tag{14}$$

where t_{m-1} and t_{m+1} are the timestamps surrounding t_m, and $x_{t_{m-1}}$ and $x_{t_{m+1}}$ are the observed values at those timestamps.

Once the data gaps are addressed, the next step is data transformation, which enriches the time series by extracting features and reducing dimensionality. Transformations like Fourier transforms convert time-domain data into the frequency domain, making it easier to detect periodic patterns that may not be obvious in the original sensor readings [48]. The discrete Fourier transform (DFT) of a time series x_t is defined as in Formula (15):

$$X_k = \sum_{t=0}^{N-1} x_t e^{-2\pi i k t / N} \tag{15}$$

where X_k is the frequency component at index k, x_t is the value of the time series at time t, and N is the total number of time points in the series.

At this stage, the preprocessed data can be used as input for models such as LSTM, which are designed to capture temporal dependencies and predict future structural health. By following these rigorous preprocessing steps, the BDT model can more effectively detect early signs of degradation, enabling timely interventions and maintenance.

4.3.3. Sensor Data Processing Using Long Short-Term Memory

LSTM networks are uniquely equipped to handle sequential data, making them ideal for analyzing time series data, where both short-term variations and long-term patterns are critical for making accurate predictions and issuing early warnings of structural issues [49].

LSTM models are an advanced variant of recurrent neural networks (RNNs), designed to address the challenge of retaining relevant information over long sequences of data [50]. Unlike standard RNNs, LSTM networks incorporate memory units that help them retain relevant information over extended time periods, enabling them to better capture patterns in time series data. The memory is regulated by several gating mechanisms, which allow the model to control the flow of information: deciding what to remember, what to forget, and what to output at each time step.

At each time step t, the LSTM updates its internal states—the hidden state h_t and the cell state C_t—based on the previous state and the current input x_t. The process is governed by the following equations.

The forget gate determines which parts of the previous cell state should be forgotten. It uses the hidden state from the previous time step h_{t-1} and the current input x_t as in Formula (16):

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right). \tag{16}$$

The input gate decides which new information should be added to the cell state. This step also considers the previous hidden state and current input as in Formula (17):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \tag{17}$$

The cell state update computes the new candidate values $\widetilde{C}_t$ for the cell state, which will be added to the existing cell state as in Formula (18):

$$\widetilde{C}_t = \tan h(W_C \cdot [h_{t-1}, x_t] + b_C). \tag{18}$$

To update the cell state, the cell state C_t is updated by a combination of the old cell state (controlled by the forget gate) and the new candidate values (controlled by the input gate) as in Formula (19):

$$C_t = f_t \times C_{t-1} + i_t \times \widetilde{C}_t. \tag{19}$$

The output gate determines what the next hidden state h_t should be, based on the updated cell state and current input, as in Formulas (20) and (21):

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{20}$$

$$h_t = o_t \times \tan h(C_t) \tag{21}$$

where f_t is the forget gate, responsible for discarding irrelevant information from the previous cell state; i_t is the input gate, which selects new information to add to the cell state; $\widetilde{C}_t$ is the candidate cell state, representing new data that could potentially update the memory; o_t is the output gate, determining the information to pass on to the next time step; σ represents the sigmoid activation function; and $\tan h$ is the hyperbolic tangent function used to normalize the updates.

Each gate plays a crucial role in regulating the flow of information, ensuring that the LSTM can retain significant historical data while forgetting less relevant details. This balance between memory retention and forgetting makes LSTM particularly effective for time series data, where patterns evolve over time but may also require attention to short-term fluctuations. For example, by analyzing previous readings of strain or displacement, an LSTM model can forecast future sensor values. Such predictions provide crucial insights into how the structure might respond to environmental changes or increased loads, allowing engineers to anticipate potential issues such as material fatigue, deformation, or abnormal stress accumulation.

One of the key advantages of employing LSTM models in the BDT model lies in their capacity to generate early warnings by analyzing temporal patterns in sensor data. By continuously evaluating the outputs from multiple sensors and comparing them with predicted values, the model can identify deviations that may indicate structural deterioration. When unexpected variations, such as sudden increases in stress or displacement, are detected, the LSTM model generates an alert, prompting immediate investigation and further analysis.

This early warning mechanism is crucial for mitigating the risk of structural failures. By detecting early signs of damage or stress, LSTM models enable maintenance teams to undertake corrective measures before the situation escalates. Such a proactive approach helps prolong the lifespan of the structure while simultaneously reducing the likelihood of sudden and catastrophic failures.

4.4. Expert Knowledge

In the DTM of bridges, in addition to relying on geometric spatial data and sensor data, expert knowledge of bridge engineering can also be integrated. Expert knowledge refers primarily to the understanding and practical experience regarding bridge structures that experts have gained through years of professional experience. This knowledge can effectively compensate for the limitations of data-driven models in evaluating complex bridge structures, providing more comprehensive support for monitoring, diagnosis, and maintenance. This section discusses the methods of collecting and processing expert knowledge.

4.4.1. Acquiring Expert Knowledge

Unlike traditional data, expert knowledge is mainly acquired through expert interviews, expert workshops, national standards, and historical case analyses. Expert interviews are a direct way to acquire expert knowledge.

Through structured interviews, knowledge of specific damage types such as bridge cracks, corrosion, and deformations can be systematically collected from experts. By systematically recording the experts' evaluation criteria and qualitative judgments for different types of damage, this information forms a key foundation for the expert knowledge base.

Expert workshops bring together multiple experts in bridge engineering to discuss particular types of bridge damage. Through discussions, a consensus is reached regarding the evaluation and assessment of the damage types. Expert workshops primarily focus on complex or controversial issues in order to form a systematic framework for damage evaluation. National standards and industry regulations are authoritative sources of expert knowledge. The standards published by the government and industry detail the methods for monitoring and assessing bridge damage under various circumstances. By carefully analyzing the content of these standards, key indicators and criteria related to bridge damage are extracted. Standards represent the primary source of data informing expert knowledge. Historical case analysis provides expert knowledge that has been validated through real-world practice. By analyzing successful cases of bridge monitoring and maintenance, specific and effective measures can be summarized. Furthermore, through historical case analysis, the development patterns and characteristics of similar types of damage, such as regions where cracks tend to concentrate or deformation trends, are identified. Historical cases provide valuable empirical insights and further enhance expert knowledge.

4.4.2. Processing of Expert Knowledge

To ensure that expert knowledge can be effectively applied in the DT system, the collected expert knowledge must be standardized and systematized.

As experts may use inconsistent terminology and standards in their assessments, the collected expert knowledge cannot be applied directly. It must first be standardized to ensure a uniform representation of knowledge. For example, concerning the "crack width" assessment indicator, a unified quantitative standard must be established to categorize the severity as "minor", "moderate", or "severe". Standardization ensures that all expert knowledge is processed consistently during later monitoring and evaluation phases.

Expert judgment is often accompanied by uncertainty and ambiguity, particularly in assessing complex structural damage. To effectively manage this expert knowledge, the system introduces fuzzy logic [51]. Fuzzy logic uses membership functions to convert qualitative judgments from experts into quantitative data, enabling the system to flexibly handle complex assessments. For instance, concerning "crack width", multiple fuzzy sets can be defined (e.g., "minor crack", "moderate crack", and "severe crack"). Each crack width is mapped to the appropriate fuzzy set through a membership function, thereby quantifying the crack severity. The system then uses the membership degrees for assessment and reasoning, providing a specific evaluation of the damage severity.

The standardized expert knowledge is used to build an expert knowledge base, which is a system for the structured storage and management of expert knowledge. Through the systematization and integration of expert experience and judgments into the DTM, the expert knowledge base classifies knowledge according to the different structural components of the bridge (e.g., piers, decks, and supports) and includes information on damage types, causes, and monitoring methods.

To further enhance the decision-making capabilities of the DTM, this expert knowledge base harmonizes qualitative insights with quantitative data from sensors and spatial models. This integration ensures a cohesive analysis of structural health, reducing fragmentation and supporting actionable insights for monitoring and maintenance.

To ensure the continuing suitability of the expert knowledge base, it must be dynamically updated based on the latest bridge monitoring cases, technological advances, and standards. By verifying and calibrating the rules within the knowledge base, its suitability across diverse bridge structures and environmental conditions is enhanced.

5. Bridge Digital Twin Modeling Using Multi-Source Heterogeneous Data

The BDT model driven by both data and expert knowledge is crucial for effectively monitoring and predicting the structural health of bridges. The BDT model consists of a spatial model, a temporal model, and an expert knowledge model. By integrating 2D image data, 3D point cloud data, and sensor data, the spatial and temporal models are established, and the expert knowledge model is developed using an expert knowledge base. Figure 8 illustrates the architecture of the BDT model, showcasing the integration of data-driven and knowledge-driven approaches for structural health monitoring. This approach facilitates a comprehensive assessment of structural health, providing a solid foundation for predicting potential risks and issuing timely early warnings.

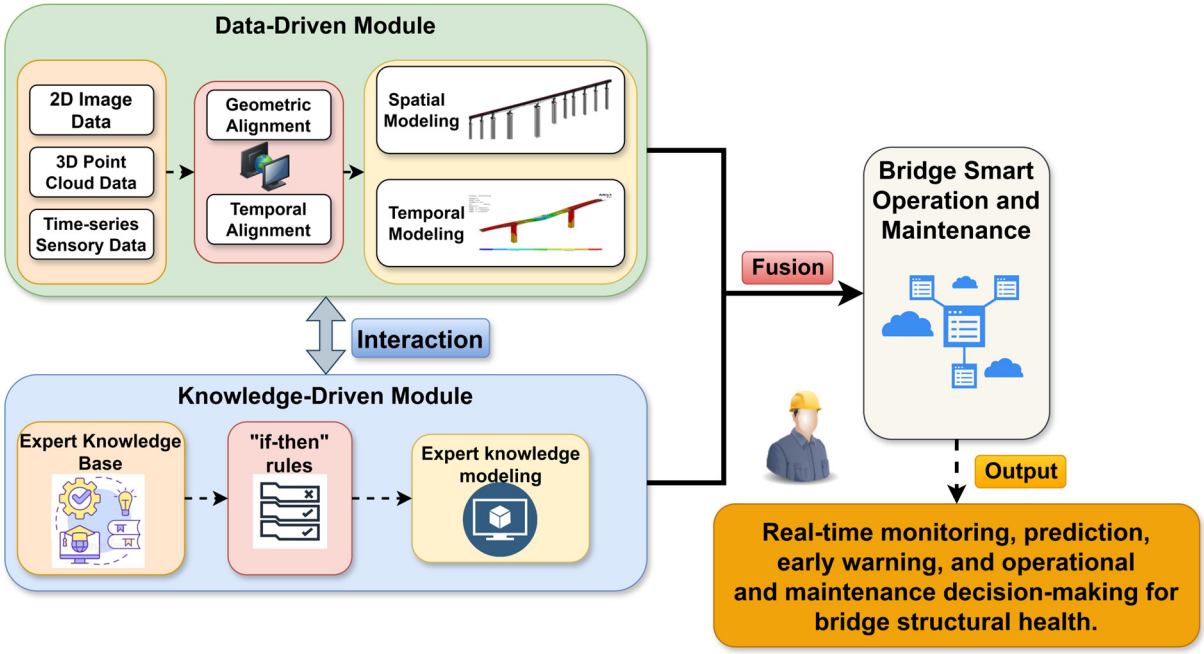


Figure 8. Integration of data and knowledge-driven modules for smart bridge operation and maintenance.

5.1. Spatial Modeling of Bridge Digital Twin

The spatiotemporal alignment of various bridge data sources is critical for (1) establishing correlations among various bridge defects and deformations and (2) explaining root causes of the identified structural health issues. Hence, the spatial modeling process begins with the spatiotemporal alignment of 2D image data, 3D point cloud data, and sensor data into a unified coordinate system, forming the foundation for spatial modeling of the bridge digital twin. Given that the spatial coordinates of sensor data are relatively straightforward to obtain, the critical challenge lies in aligning the 2D image data with the 3D point cloud data, which is key to achieving precise spatial modeling.

The correspondence between 2D image data and 3D point cloud data can be established using the speeded-up robust features (SURF) algorithm, which extracts key feature points [52]. This is followed by initializing transformation matrices and parameters, with refinement achieved through the iterative closest point (ICP) algorithm to enhance alignment accuracy.

First, SURF is applied to extract key feature points from the 2D image data. For faster computation, SURF employs the integral image method, where the integral image is defined as in Formula (22):

$$I_{integral}(x, y) = \sum_{i=0}^{x} \sum_{j=0}^{y} I(i, j) \tag{22}$$

where $I(i,j)$ represents the pixel intensity at point (i,j), and $I_{integral}(x,y)$ is the cumulative sum of pixel intensities from the top-left corner $(0,0)$ to (x,y).

SURF detects key points using the Hessian matrix, defined as in Formula (23):

$$H(x,y,\sigma) = \begin{bmatrix} L_{xx}(x,y,\sigma) & L_{xy}(x,y,\sigma) \\ L_{xy}(x,y,\sigma) & L_{yy}(x,y,\sigma) \end{bmatrix} \tag{23}$$

where L_{xx}, L_{xy}, and L_{yy} are second-order partial derivatives in the x and y directions, and σ represents the scale. The key feature points are identified based on the response of the Hessian matrix. Subsequently, feature points from the 2D image and geometric feature points from the 3D point cloud data are extracted and matched to form corresponding pairs. The next step involves initializing the rotation matrix R and the translation vector T between the two datasets. A least-squares method is then applied to solve for the transformation matrix, ensuring that the transformed 2D image data align as closely as possible with the corresponding points in the 3D point cloud data as in Formula (24):

$$\min_{R,T} \sum_{i=1}^{N} \| p_i^{3D} - \left(R \cdot p_i^{2D} + T \right) \|^2 \tag{24}$$

where p_i^{3D} represents the corresponding point in the 3D point cloud data, while p_i^{2D} denotes the feature point in the 2D image data. N refers to the total number of matching points. The initial alignment is achieved by calculating the initial rotation matrix R_0 and translation vector T_0 using singular value decomposition. The alignment is then iteratively refined using the iterative closest point (ICP) algorithm. This process continuously reduces the distance between the two point sets, progressively optimizing the rotation matrix and translation vector. The iteration continues until the error converges to a predefined threshold or the maximum number of iterations is reached, yielding the final optimized rotation matrix R and translation vector T.

Upon completing the geometric alignment of the 2D image data and 3D point cloud data, the geometric alignment of the sensor data is accomplished by measuring the actual installation locations of the sensors to obtain their spatial coordinates. These coordinates are then used to map the sensor data into the unified coordinate system.

5.2. Temporal Alignment of Bridge Digital Twin

The temporal alignment of 2D image data, 3D point cloud data, and sensor data is crucial for temporal modeling. By achieving geometric and temporal alignment of multidimensional data, critical data support can be provided for the DTM of bridges. In the modeling of multi-source heterogeneous data fusion, the resampling method is used to align data in time. Since the sampling times and frequencies of 2D image data, 3D point cloud data, and sensor data vary, resampling involves selecting an appropriate sampling frequency to achieve time alignment.

Selecting the appropriate sampling frequency is crucial during the resampling process. Sensor data typically have a higher sampling frequency than 2D image data and 3D point cloud data, which are captured at lower frequencies. Therefore, the sensor data sampling rate must be reduced to synchronize it with the acquisition frequency of the 2D image and 3D point cloud data. By establishing an appropriate sampling frequency, the timestamps of the sensor data can be aligned with those of the other data sources, ensuring consistent temporal synchronization across datasets.

Sensor time series data often contain a significant amount of high-frequency information, which can accumulate in large datasets during long-term bridge health monitoring.

Resampling not only retains the essential physical information but also reduces unnecessary data accumulation, thereby accelerating the data processing speed of the DTM.

5.3. Expert Knowledge Modeling and Integration

Expert knowledge modeling plays a vital role in the DTM of bridges. It is not merely the formalization of rules from the expert knowledge base; more significantly, it integrates with spatial and temporal models to create a comprehensive evaluation system for more accurate and holistic assessments.

The expert knowledge model establishes a rule engine, which is applied to real-time data and predictive trends derived from spatial and temporal models. Based on different input conditions, such "if–then" rules enable damage assessment and risk prediction. For example, "If the crack width surpasses 5 mm, the structural integrity may be at risk".

The expert knowledge model also incorporates a data linkage engine based on the rule engine and expert experience. When data from the spatial and temporal models are integrated into the expert knowledge model, it combines expert experience with multi-source data around the anomaly points to assess the condition of the bridge. This allows for a more comprehensive evaluation of the bridge's state by considering anomalies in context with nearby data.

The expert knowledge model integrates the rule engine and data linkage engine with expert knowledge to analyze and predict the types and locations of potential damage to the bridge. First, the rule engine detects potential anomaly points in the sensor data, triggering system alerts. Then, the data linkage engine further examines these anomalies by incorporating historical data, environmental data, and inputs from the spatial and temporal models to precisely locate the problem and predict damage development in that area.

The expert knowledge model dynamically adjusts the rules according to expert knowledge in different environments. For example, in extreme temperatures or high-humidity conditions, the aging process of certain bridge materials may accelerate significantly. The expert knowledge model accounts for the influence of environmental factors and updates the rules to address new damage patterns or accelerated damage progression. The dynamic adjustment of rules based on expert knowledge enables the model to respond with greater flexibility and precision to real-time data.

5.4. Establishing Bridge Digital Twin for Smart O&M

After completing spatial modeling, temporal modeling, and expert knowledge modeling, the next crucial step is integrating these models to finalize the DTM of the bridge. This process consolidates different models, where spatial and temporal models serve as data inputs, and the expert knowledge model functions as an evaluation system that assesses the structural health of the bridge.

The 2D image data primarily capture visual information and are often used to monitor surface damage such as corrosion and cracks. The 3D point cloud data provide accurate spatial information, primarily used to monitor overall structural deformations and localized displacements. Sensor data capture the physical state of the bridge, encompassing parameters like stress, strain, and temperature. In this process, the expert knowledge model, functioning as a rule-based system, can identify areas of concentrated stress, allowing for further evaluation of critical regions.

By combining data with expert knowledge for comprehensive analysis, integrated monitoring of the bridge's health across spatial and temporal dimensions can be achieved. For example, surface damage detected by 2D images can be precisely located using 3D point cloud data, while sensor data at that location can supplement the physical information. Similarly, the expert knowledge model can further analyze areas of concentrated stress

identified in the sensor data, providing more detailed assessments. Furthermore, the temporal model and expert knowledge model can predict the future condition of the bridge, enabling early warnings. By applying LSTM deep learning models to the sensor time series data, trends in stress changes within the bridge can be forecasted [50].

For instance, the predicted development of stress concentrations combined with changes in 2D image and 3D point cloud data can help predict the progression of cracks or deformations. When these predictions are further evaluated using the expert knowledge model, risk areas can be identified earlier, enabling timely maintenance and repairs, thereby minimizing the risk of sudden structural failures. Additionally, the three models can be used for cross-validation to enhance the accuracy and reliability of bridge monitoring. For example, abnormal deformations detected by 3D point cloud data can be validated by checking the corresponding strain data from sensors or referring to historical cases and experiential knowledge in the expert knowledge model. This cross-validation strengthens the robustness of the monitoring system and reduces the likelihood of errors from any single data source.

In conclusion, the data and expert knowledge-driven DTM of the bridge provides comprehensive support for SHM. This approach not only detects surface damage but also tracks geometric deformations and physical stress conditions. Furthermore, through time series analysis, it predicts future trends and performs grading assessments using the expert knowledge model. These technologies enable more effective assessments of the bridge's health and improve the accuracy of fault detection, ultimately enhancing the safety and reliability of the bridge.

This study evaluated the proposed method through its application to the Hedong Bridge, located in Anhui Province, China. The Hedong Bridge, spanning 612 m, is a prestressed concrete continuous girder bridge, selected for its structural significance and suitability for validating advanced methodologies. The experimental process commenced with the acquisition of 2D image data using UAVs equipped with high-definition cameras, which captured detailed surface imagery along predefined flight paths. Concurrently, UAVs with laser scanners collected 3D point cloud data, providing precise geometric information. In addition, sensor arrays—comprising strain gauges, accelerometers, temperature sensors, and GNSS positioning devices—were strategically deployed at critical locations on the bridge to record time series data of its physical properties. To ensure data accuracy and minimize environmental interferences, the UAV flight paths and sensor configurations were carefully calibrated. Following data collection, rigorous preprocessing was conducted to enhance data quality. The processed datasets were analyzed using advanced computational models tailored to each data type. The YOLOv8 algorithm was employed to detect surface defects, accurately identifying cracks and spalling in multiple locations. The PointNet model analyzed the 3D point cloud data, revealing structural deformations, including a vertical displacement of 12 mm at the mid-span. The LSTM model processed time series sensor data, predicting vibration amplitude trends and identifying potential fatigue under traffic loads. These processed datasets were integrated into the DK-DTM framework and further enriched with an expert knowledge model, incorporating historical maintenance records and design specifications for the Hedong Bridge. This comprehensive integration facilitated an in-depth assessment of the bridge's health risks and generated precise maintenance recommendations. The proposed framework effectively identified high-priority areas requiring immediate repair and confirmed the presence of early-stage fatigue risks. This case study of the Hedong Bridge substantiated the feasibility and reliability of the proposed methodology in addressing critical challenges in structural health monitoring. The detailed experimental workflow is depicted in Figure 9.

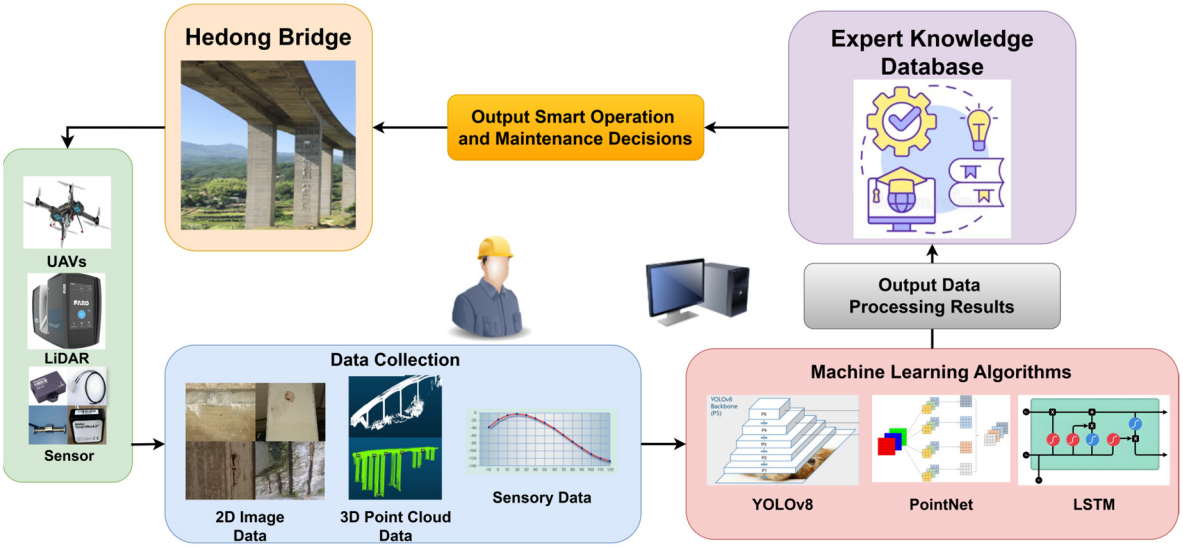


Figure 9. Experimental process for the application of the DK-DTM framework to the Hedong Bridge.

6. Discussion

This study proposed a DTM for bridges that integrates spatial, temporal, and expert knowledge models. This model, jointly driven by data and expert knowledge, enables precise monitoring and prediction of bridge structural health. In comparison with traditional DTM, this study's proposed model offers enhanced proactive state assessment capabilities, thereby advancing the development of smart O&M systems.

In contrast to prior studies, this paper modeled data separately in spatial and temporal dimensions. The spatial model recorded real-time bridge state data, while the temporal model forecasted trends in bridge state data. Furthermore, integrating an expert knowledge model enhanced the DTM's logical reasoning capabilities, allowing for the assessment and prediction of bridge health conditions. Through the establishment of a data-driven module and an expert knowledge evaluation module, the model's proactive capability in bridge health monitoring was significantly enhanced.

The proposed DTM improves the bridge maintenance process in several key ways. First, by leveraging real-time data acquisition, the model continuously monitors bridge health, allowing early detection of potential issues. The temporal model further analyzes data trends to predict future changes in bridge deterioration at both surface and internal levels. Additionally, integrating an expert knowledge model enhances logical reasoning, enabling more efficient monitoring and prediction of bridge status. Leveraging bridge state predictions, the model optimizes O&M workflows to reduce unnecessary inspections, improving resource allocation. This proactive approach helps prevent sudden damage and minimize long-term degradation, ultimately extending the bridge's lifespan.

Despite advancements in applying DT technology in this study, certain limitations and challenges persist. Data accuracy critically influences the overall performance of the DTM, especially in large-scale bridges, where data volumes are extensive. The integration speed of multi-source heterogeneous data affects the model's real-time performance. Additionally, managing non-deterministic and specialized bridge expert knowledge during expert knowledge base construction poses challenges. The robustness of machine learning algorithms in the model largely depends on training data quality and diversity, potentially impacting the model's generalizability across bridge types and environmental conditions.

7. Conclusions

This study proposed a novel digital twin (DT) framework that integrates spatial, temporal, and expert knowledge models to enhance real-time bridge monitoring and maintenance. By addressing critical challenges in bridge maintenance, this framework improves the accuracy of structural assessments and increases the efficiency of maintenance operations, demonstrating the proactive role of DT technology in bridge O&M.

The findings highlight the potential of DT technology to extend the lifespan of critical infrastructure through real-time data integration and predictive maintenance. The framework provides a practical solution for smart structural health monitoring and contributes to the development of advanced strategies for infrastructure management.

In summary, this study underscores the significance of DT technology in improving structural safety and operational efficiency. The proposed framework serves as a foundational step toward implementing smart, data-driven solutions in civil engineering, paving the way for further advancements in intelligent infrastructure management.

Future studies could explore several key directions to further enhance DT applications in bridge monitoring. First, incorporating advanced machine learning algorithms may improve model efficiency and accuracy. Second, improving the interaction between spatial and temporal models could increase model responsiveness, strengthening real-time decision-making. Refining expert knowledge model classifications and enabling automatic selection of appropriate knowledge bases for varying bridge types could enhance operational efficiency. Such advancements will continue to evolve DT technology, enhancing infrastructure monitoring and maintenance in intelligence, efficiency, and precision.

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